Supporting Information

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SI Materials and Methods

Materials. Fluticasone propionate, halcinonide, and clobetasol propionate were independently supplied by Prestwick Chemicals and Sigma. Fluocinonide was repurchased from Prestwick Chemicals. SAG was purchased from Axxora. Purmorphamine was purchased from Calbiochem. Cortisone, dexamethasone, prednisolone, and corticosterone were purchased from Sigma. Cyclopamine and bodipy-cyclopamine were purchased from Toronto Research Chemicals.

Transfection and Plasmids. Expression vectors for βarr2-GFP, WT Smo, Smo-YFP, and Gli-luciferase reporter have been described (1–3). V2R-GFP was made in a similar manner as β2-adrenergic receptor-GFP (4). GR-GFP was provided by Terry Hinds, Jr. (University of Toledo, Toledo, Ohio). Shh conditional medium (Shh) was prepared by collecting the medium in which HEK293 cells were transfected with Shh-N plasmid (provided by Philip Beachy, Stanford University, Stanford, CA). Smo-633 was constructed by swapping the 154 amino acids of the Smo C terminus (Pro-634 to Phe-787) with the 29 amino acids of the V2R C terminus (Ala-343 to Ser-371).

Primary Assay-Automated High-Throughput Screening. Smo-633 was used in this assay because it produces a stronger signal in the U2OS Smo/βarr2-GFP assay than WT Smo, but it is otherwise pharmacologically similar. A single 15-cm plate of U2OS cells stably expressing Smo-633 and βarr2-GFP at 95–100% cell confluency was split into two glass-bottom 384-well plates (MGB101-1-2-LG; Matrical) using a Multidrop 384 dispenser (Titertek Instruments). Each well contained 25-μL aliquots of 6,000 cells in MEM [10% (vol/vol) FBS, 50 U/mL Pen/Strep] with 100 nM cyclopamine. Plates were incubated overnight at 37 °C in 5% (vol/vol) CO_2. On the following day, compounds (5 mM in DMSO) from the Prestwick Chemicals Library and the positive control (5 mM SAG) as well as the vehicle control DMSO were diluted 1:200 in MEM, and 6.25 μL of the diluted compound, SAG, or DMSO was added to each of the corresponding wells using a Biomek FX liquid handler (Beckman Coulter), for an overall 1:1,000 dilution and final concentration of 5 μM. The plates were returned to the incubator for 2 h. The media was then removed, and the cells were fixed in 30 μL of PBS containing 0.5% paraformaldehyde and a 1:50,000 dilution of DRAQ5 dye (Biostatus) to visualize the nuclei. Plates were stored at 4 °C until analysis on an ImageXpress Ultra (Molecular Devices) equipped with a 488-nM argon laser for imaging GFP and a 568-nM krypton laser for imaging DRAQ5. Plates were screened in duplicate at the rate of 5,000 wells per day and analyzed using the associated software Transfluor HT (Molecular Devices), and results were visually confirmed.

Receptor Internalization Assay. Smo-YFP and V2R-GFP internalization assays were performed as previously described (1).

Bodipy-Cyclopamine Binding Analysis of Smo Agonists in Smo-Overexpressing HEK293 Cells. Cells overexpressing wild-type Smo were washed and fixed with 4% (vol/vol) formaldehyde/PBS for 20 min at room temperature. After removing the formaldehyde buffer, the cells were incubated for 2 h at room temperature in binding buffer (HBSS without Ca^{2+} and Mg^{2+}) containing 0–50 nM bodipy-cyclopamine for saturation binding assay or 5 nM bodipy-cyclopamine and compounds over a range of concentrations from 0–10 μM for competitive binding assay. Following incubation, the cells were washed three times with the binding buffer and imaged using a Zeiss ×40 oil N.A. 1.3 plain apo objective and 488-nm excitation laser on a Zeiss LSM-510 confocal microscope. The average fluorescence intensity corresponding to each image (i.e., concentration) was determined using the accompanying histogram software. The specific binding data over baseline for compounds were normalized to the maximal binding of bodipy-cyclopamine. The data were analyzed by GraphPad Prism (GraphPad Software, Inc.) using linear and nonlinear regression.

[^{3}H]Thymidine Proliferation Assay and Western Blots of Primary Neuronal GCPs. Briefly, cerebella were removed from 4- or 8-day-old mice, minced, and digested at 37 °C in papain before being triturated in buffer containing trypsin inhibitor and BSA. Cells were then pelleted, resuspended in BSA/PBS, and strained to remove debris. Strained cells were recentrifuged and resuspended in supplemented neurobasal medium. The cells from 8-day-old mice were cultured for 48 h in the presence of compounds with or without Shh and were then pulsed with [^{3}H]thymidine and cultured for an additional 16 h before being measured for [^{3}H]thymidine incorporation.

The cells from 4-day-old mice were cultured for 64 h in the presence of DMSO vehicle, compounds, or 2% Shh. The cells were harvested in the SDS sample buffer. The protein samples were subjected to SDS/PAGE and transferred to nitrocellulose membranes, followed by incubation with three primary antibodies: cyclin D2 and cleaved caspase 3 (Cell Signaling) and actin (Santa Cruz Biotechnology, Inc.). Subsequently, immunoblots were developed using HRP-conjugated antibodies.

1. Chen W, et al. (2004) Activity-dependent internalization of smoothened mediated by beta-arrestin 2 and GRK2. *Science* 306:2257–2260.
2. Taipale J, et al. (2000) Effects of oncogenic mutations in Smoothened and Patched can be reversed by cyclopamine. *Nature* 406:1005–1009.
3. Chen JK, Taipale J, Young KE, Maiti T, Beachy PA (2002) Small molecule modulation of Smoothened activity. *Proc Natl Acad Sci USA* 99:14071–14076.
4. Barak LS, et al. (1997) Internal trafficking and surface mobility of a functionally intact beta2-adrenergic receptor-green fluorescent protein conjugate. *Mol Pharmacol* 51:177–184.

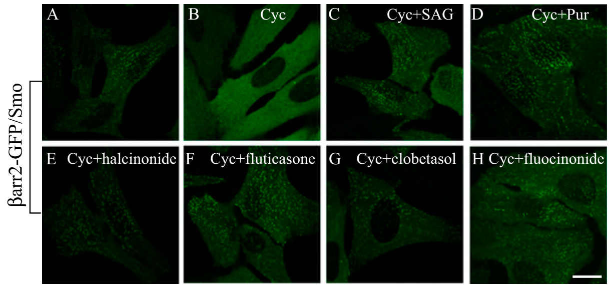

Fig. S1. Fluorinated glucocorticoid Smoothened agonist (FGSA) drugs halcinonide, fluticasone, clobetasol, and fluocinonide as well as cyclopamine, SAG, and purmorphamine regulate the membrane distribution of βarr2-GFP in cells stably overexpressing WT Smo and βarr2-GFP. (*A–H*) Confocal images of βarr2-GFP stably expressing Smo and βarr2-GFP in U2OS cells. Cells were left untreated (*A*) or treated with 100 nM cyclopamine (*B*), 100 nM cyclopamine and 5 μM SAG (*C*), 100 nM cyclopamine and 5 μM purmorphamine (*D*), 100 nM cyclopamine and 5 μM halcinonide (*E*), 100 nM cyclopamine and 5 μM fluticasone (*F*), 100 nM cyclopamine and 5 μM clobetasol (*G*), or 100 nM cyclopamine and 5 μM fluocinonide (*H*) for 2 h at 37 °C. In contrast to HEK293 cells, which are thicker and show aggregates at the periphery in confocal images [Chen W, et al. (2004) Activity-dependent internalization of smoothened mediated by beta-arrestin 2 and GRK2. Science 306:2257–2260], the flatter U2OS cells show aggregates over the whole surface. Representative images of three independent experiments are shown. Cyc, cyclopamine; Pur, purmorphamine. (Scale bar: 10 μm.)

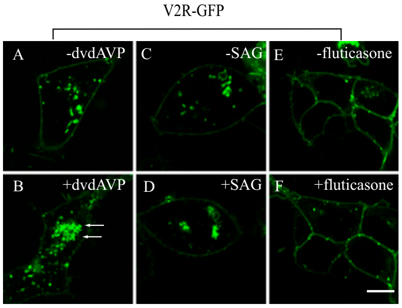

Fig. S2. SAG and fluticasone have no effects on V2R-GFP internalization. Confocal images of V2R-GFP expressed in HEK293 cells left untreated (*A*, *C*, and *E*) or treated with 2 μM V2R agonist 1-deamine-4-valine-D-arginine vasopressin (dvd AVP) (*B*), 2 μM SAG (*D*), and 2 μM fluticasone (*F*) for 30 min at 37 °C. Representative images of three independent experiments are shown. Arrows indicate internalized V2R-GFP. (Scale bar: 10 μm.)

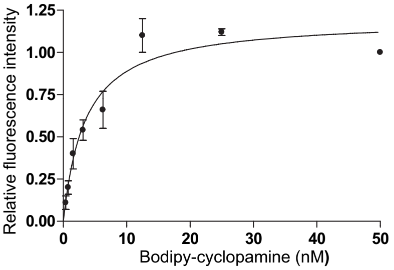

Fig. S3. Bodipy-cyclopamine saturation binding to overexpressed Smo in HEK293 cells. Binding assays were performed as described in *Materials and Methods*. The affinity (K_d) of cyclopamine for Smo was determined as 3.5 ± 0.8 nM using the program GraphPad Prism (GraphPad Software, Inc.). Data were acquired in triplicate from three independent experiments and are presented as the mean ± SEM.

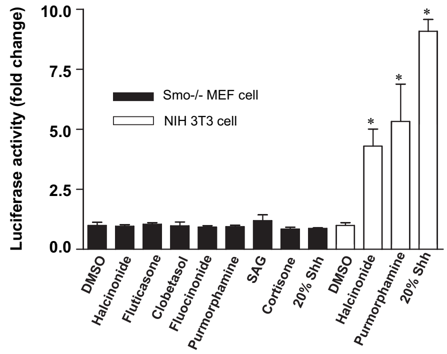

Fig. S4. Signaling of Smo agonists in $Smo^{-/-}$ mouse embryo fibroblast (MEF) cells and in NIH 3T3 cells. $Smo^{-/-}$ MEF cells or NIH 3T3 cells were transfected with Gli-luciferase reporter and control renilla pRL-TK (Promega) plasmids. The cells were cultured to confluence and treated with DMSO, 5 μM of the indicated compounds, or 20% Shh for 30 h. In $Smo^{-/-}$ MEF cells, none of the treatments resulted in a Gli-luciferase response over baseline. In NIH 3T3 cells, Gli-luciferase activities were induced by halcinonide, purmorphamine, or 20% Shh. Data are from three individual experiments performed in triplicate and presented as the mean ± SEM. The statistical significance was analyzed by a two-tailed Student's *t* test, with $^*P < 0.05$ ($\alpha = 0.05$) defined as significant (over DMSO).

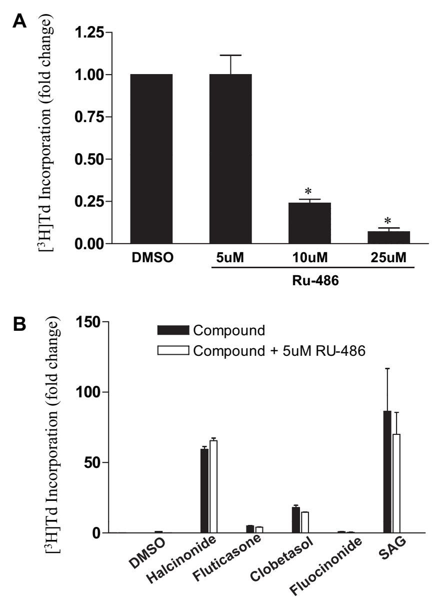

Fig. S5. GCP proliferative response to Smo agonists in the presence of mifepristone (RU-486). (*A*) RU-486 alone and GCP proliferation are shown. Primary neuronal GCPs were treated for 48 h with DMSO or RU-486 (5, 10, and 25 μM), pulsed with $[^3H]$thymidine ($[^3H]$Td), cultured for 16 h, and then assessed for $[^3H]$Td incorporation. RU-486 inhibited GCP proliferation in a dose-dependent manner. RU-486 at 5 μM had no effect on GCP proliferation. (*B*) Effect of RU-486 on Smo agonist-stimulated GCP proliferation. Primary neuronal GCPs were treated with DMSO, 12.5 μM halcinonide, 2.5 μM fluticasone, 25 μM clobetasol, 25 μM fluocinonide, or 0.5 μM SAG in the presence and absence of 5 μM RU-486. RU-486 at 5 μM had no effect on Smo agonists tested. Data obtained in triplicate are presented as the mean ± SEM ($n = 3$). The statistical significance was analyzed by a two-tailed Student's *t* test, with $^*P < 0.05$ ($\alpha = 0.05$) defined as significant (over DMSO).

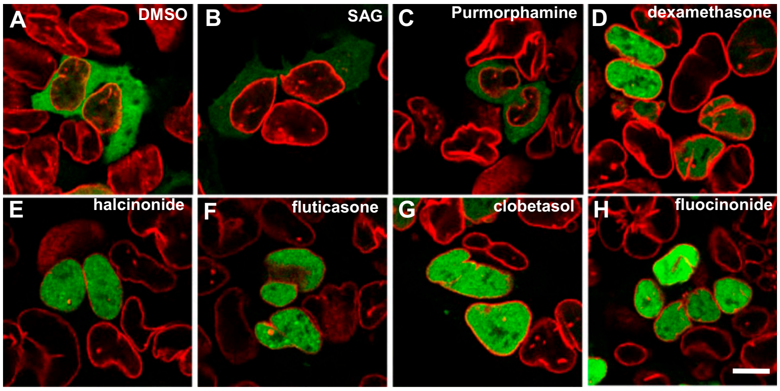

Fig. S6. Halcinonide, fluticasone, clobetasol, fluocinonide, and other glucocorticoids activate glucocorticoid receptor-GFP (GR-GFP) as assessed by GR-GFP nuclear translocation. GR-GFP-transfected HEK293 cells were treated with the indicated compound at a dose of 1 μM for 1 h, followed by immunostaining of endogenous Lamin B1 protein (red) to visualize nuclear membrane. DMSO, SAG, or purmorphamine treatment did not cause translocation of GR-GFP from the cytosol to the nucleus. Halcinonide, fluticasone, clobetasol, or fluocinonide, as well as dexamethasone, induced translocation of GR-GFP from the cytosol to the nucleus. (Scale bar: 5 μm.)